Part B : Convergent Thinking

Creating accessible kitchen utensils by re-purposing kinetic energy

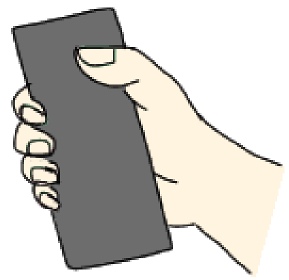
Elysia Williams

Initial sketch analysis of the Comfortable Cookie-Cutter

Taking into account this sketch analysis of my initial design from Part A , going forward I have decided to expand the range of attachments available to include a potato mashing attachment. This will give the product more versatility and therefore there are more opportunities to re-purpose wasted kinetic energy in the kitchen.

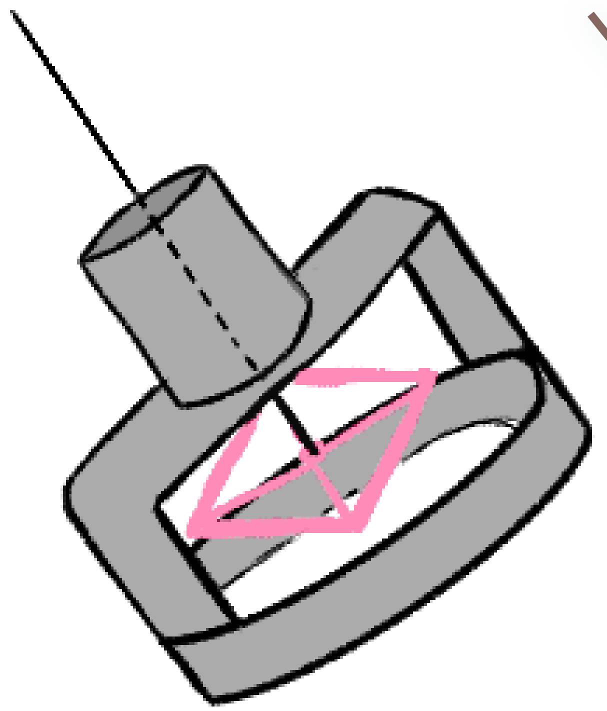
The handle is a large diameter cylinder, as explored in Part A.

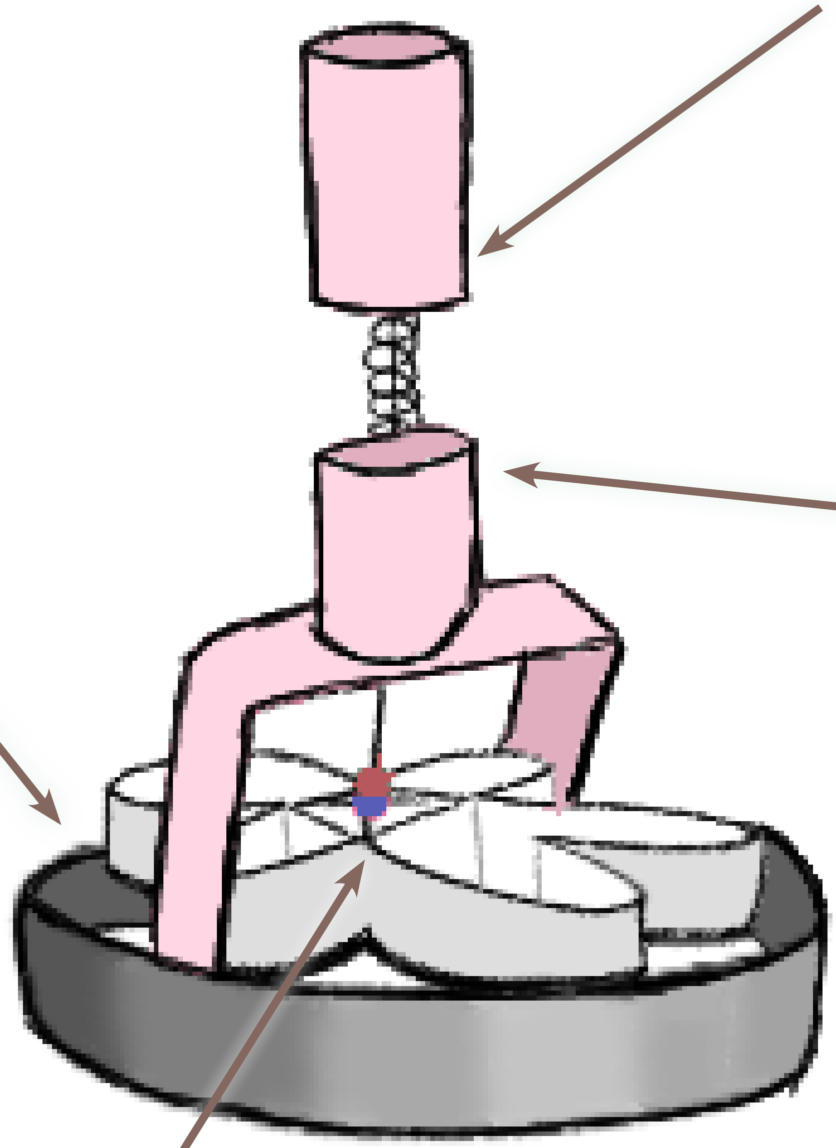
It is made of a malleable material which moulds to the user's hand.



The cutter is initially raised above the guide.

When the handle is fully pushed down, the bottom of the cookie-cutter should be level with the bottom of the guide.



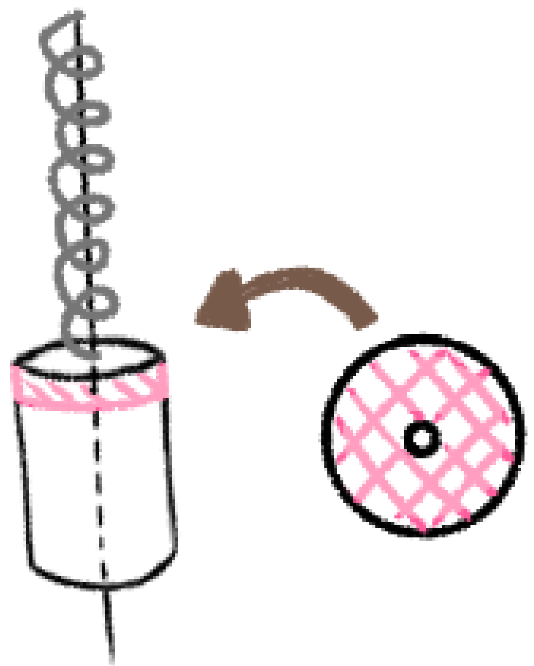


Piezoelectric material underneath the spring.

Strain is applied, generating electricity.

This is used to heat up the handle

Seed-planting attachment is outside the kitchen domain

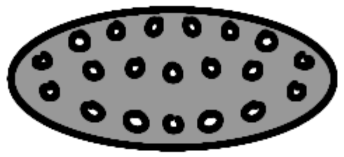


Could there be a cookie cutter that could mould to the shape the user wants?

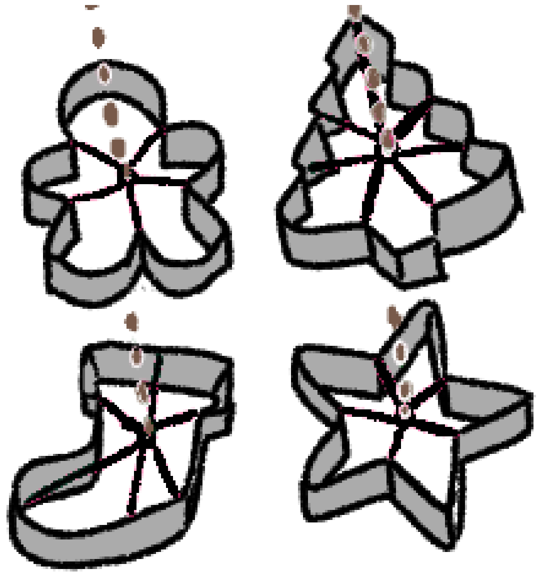
A variety of cookie-cutter shapes can be attached to the pin to be used.

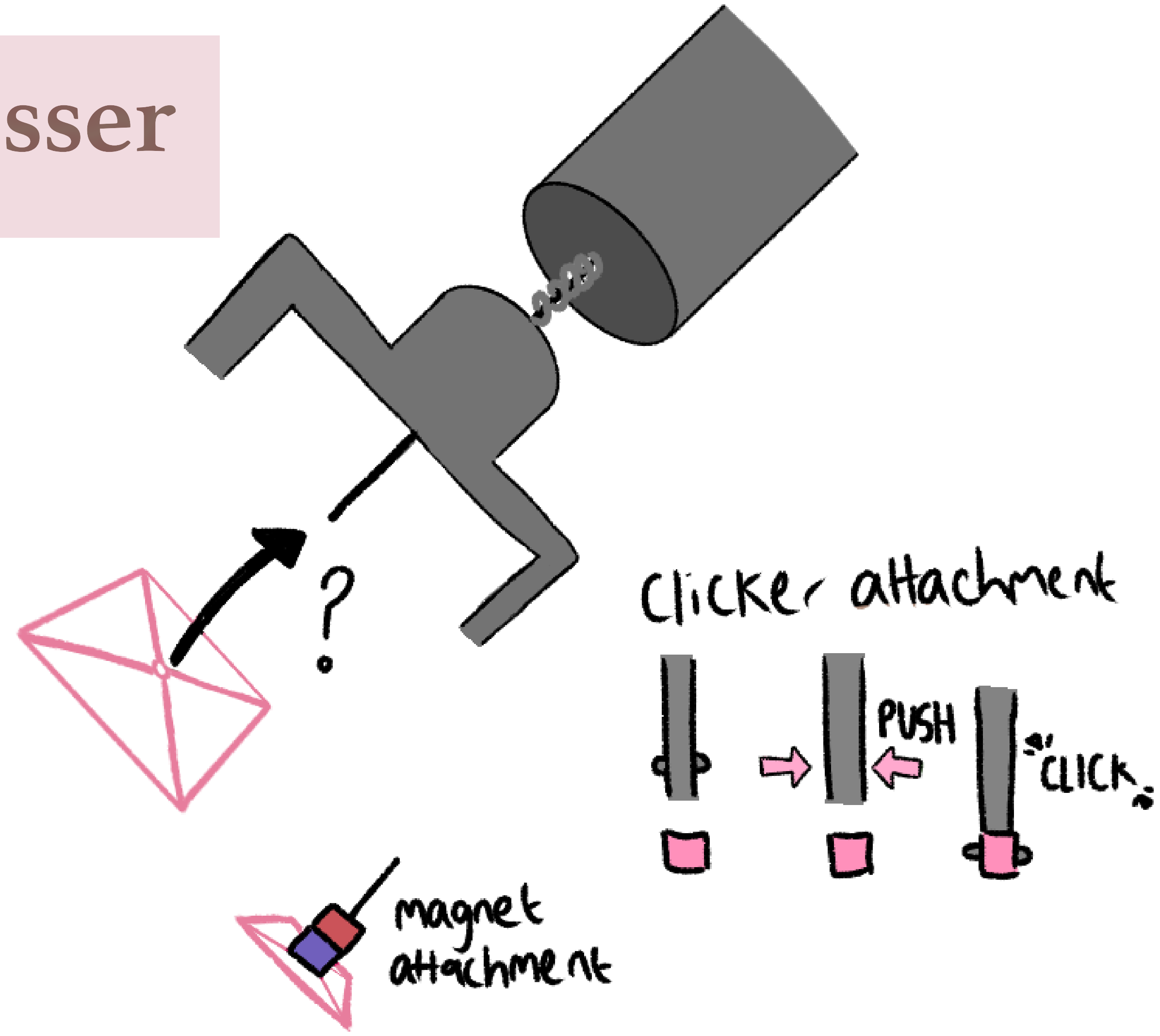
Could there be a potato masher attachment, or even a seed-planting attachment?

Requirements analysis - Accessible Presser



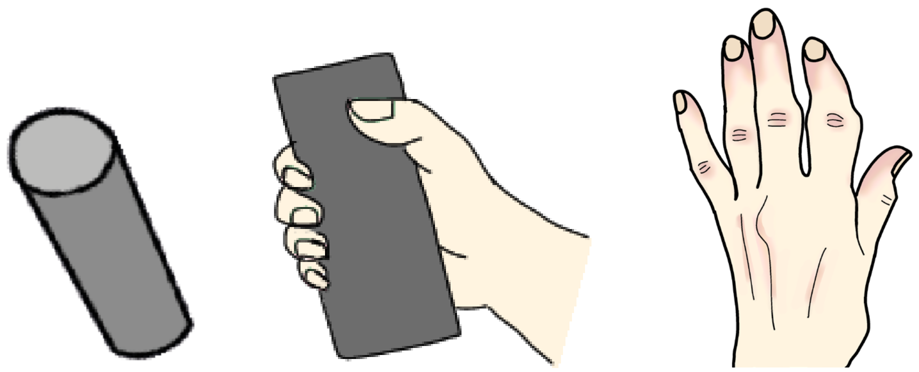
It was decided that a mould-able cookie cutter would not be suitable for someone with arthritis because they would struggle moulding it or it could fail to cut the dough - cookie shapes must not be malleable.

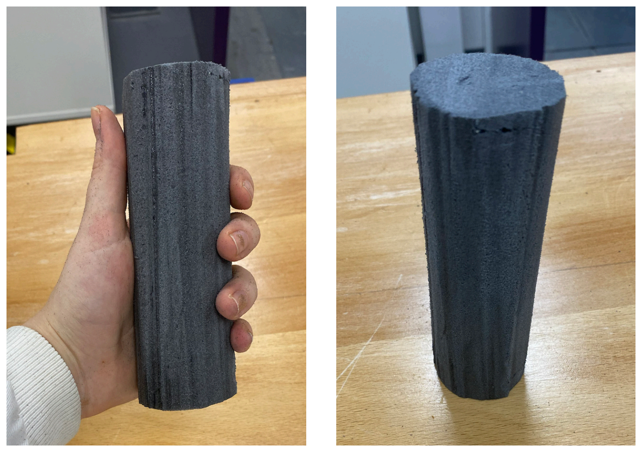




The presser must have a range of different attachments available that can be connected to the pin: cookie shapes and a potato masher attachment.

The pin must be able to attach to a range of different attachments.
This could be achieved by a magnet attachment or a clicker attachment.





Handle must be malleable and comfortable to hold.

Handle must be easy to grip. Needs to be suitable for big and small hands.

Suitable for a user with arthritis.

Through lofi prototyping in Part A, it was decided that the most comfortable shape to hold would be a large diameter cylinder

The edges of the cookie-cutter attachments must not be sharp.

Brief and Specifications

More than 10 million people in the UK have arthritis or similar conditions. Arthritis attacks the tissue in your joints, causing swelling and stiffness. It is very common in the joints of the hand. This leads to increased difficulty in performing daily tasks, such as cooking. There is also a lot of kinetic energy wasted during daily tasks, particularly in the kitchen.

The Accessible Presser could help people with arthritis to feel more at ease when cooking.

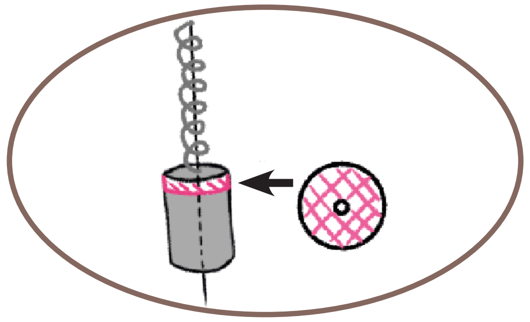
The aim of this project is to create a more accessible kitchen utensil that uses re-purposed kinetic energy to aid those with arthritis or decreased dexterity.

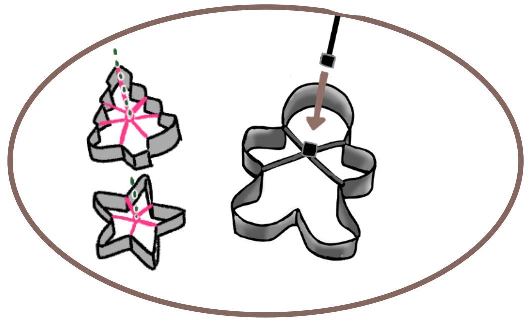
Rank	Requirement	Specification
1	Handle must heat up through re-purposed kinetic energy	Piezoelectric material in the presser that re-purposes kinetic energy into electrical energy.
2	Malleable while elastic handle so it doesn't fracture under repeated reshaping	Memory foam handle – moulds to the user's hand and springs back after use.
3	Handle must correlate to the anthropometric data of palm width.	The 95th percentile for men is 9.76. The diameter of the cylinder should be 9.76 cm so it is not too small.
4	Pin must attach to a range of cookie-cutter shapes and the potato masher attachment.	Attachments can be connected to the pin through a magnetic or clicker attachment.
5	Attachments must be sturdy but safe, cookie-cutter must not cut flesh	Rounded edges
6	Rust resistant spring	Stainless steel spring

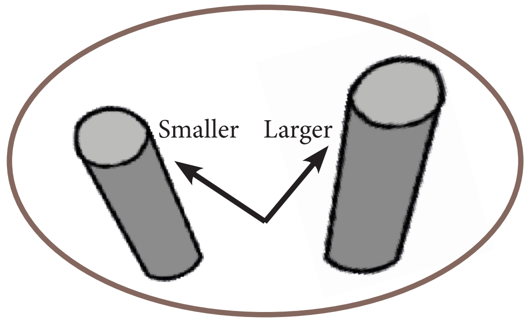
Morphological Analysis

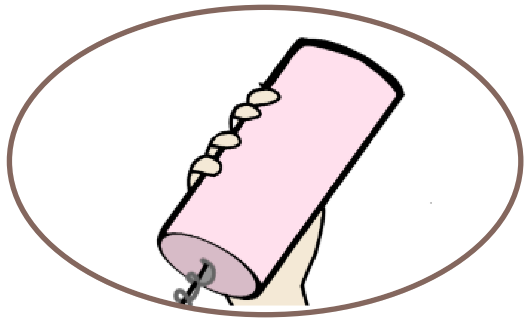
Concepts

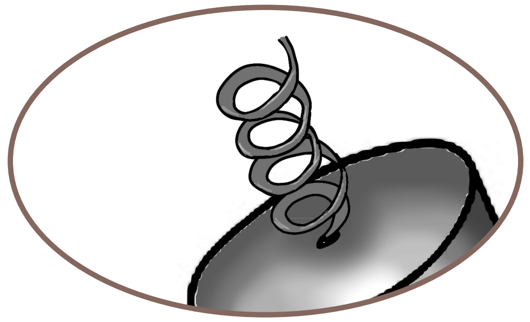
Function	1	2	3	4
Handle must heat up	Piezoelectric material generates electricity	Battery operated	Turbine operated by airflow through the handle	
Pin must attach to a range of shapes	Magnetic attachment	Suction attachment	Clicker attachment	
Handle must fit comfortable in the hand, not too small or it can irritate the joints, but not too big	Only use the 95th percentile palm width for men, since it is the biggest, the handle will not be too small.	Separate handles sold for 95th percentile palm width of men and women.		
Malleable while elastic handle that doesn't fracture under repeated reshaping	Memory foam	Thermoplastic	Silicone	
Rust resistant steel spring	Nickel plating	Copper plating	Chromium plating	Zinc plating
Cookie cutter must not cut flesh	Softer silicone cutter	Rounded edge metal cutter	Hard plastic cutter	

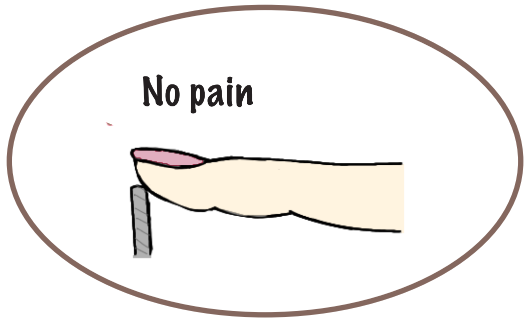






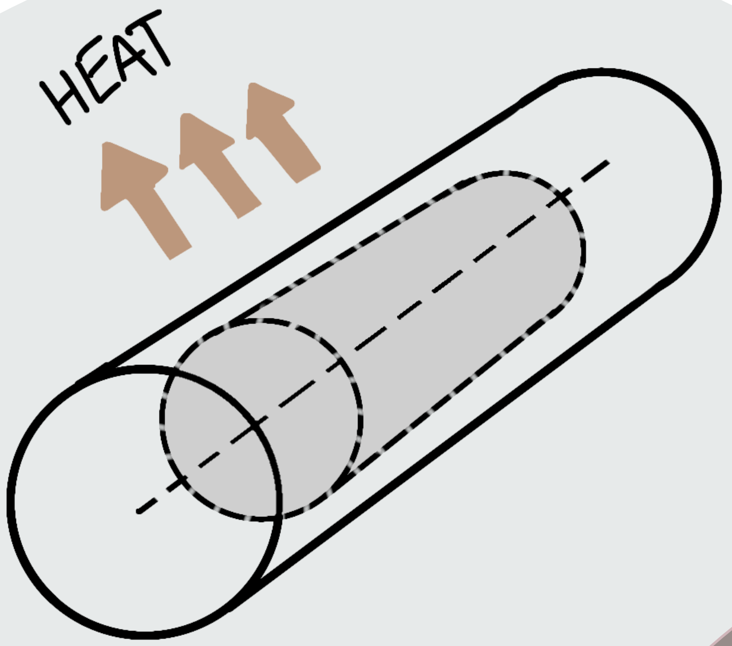






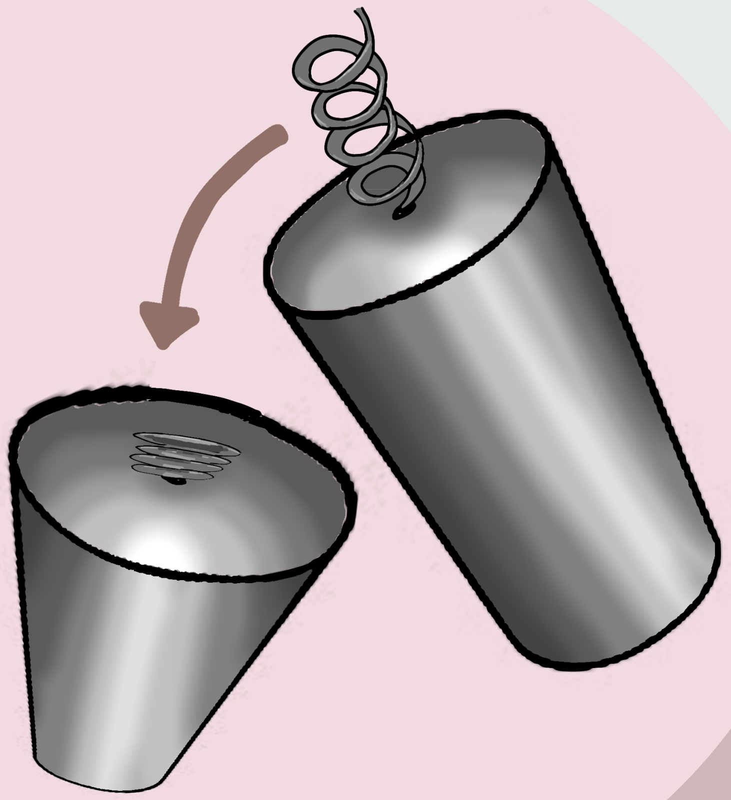
Enhanced sketching

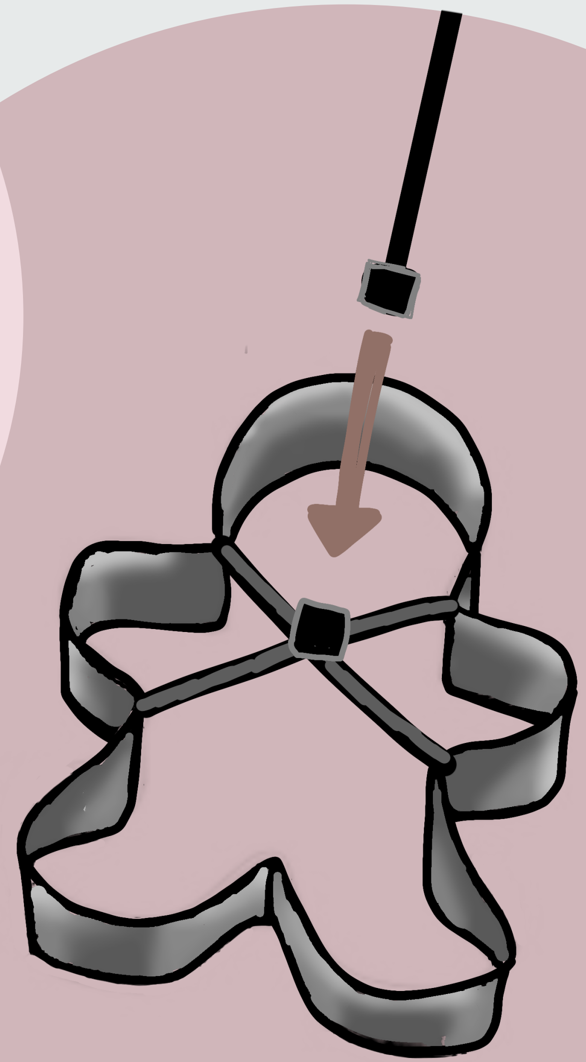
As the spring is compressed down, strain is applied to the piezoelectric material,
converting kinetic energy to electrical energy. This is used to power the heater in the handle.



Sketching arthritic hands to explore what changes must be made to suit the structure of their hands.

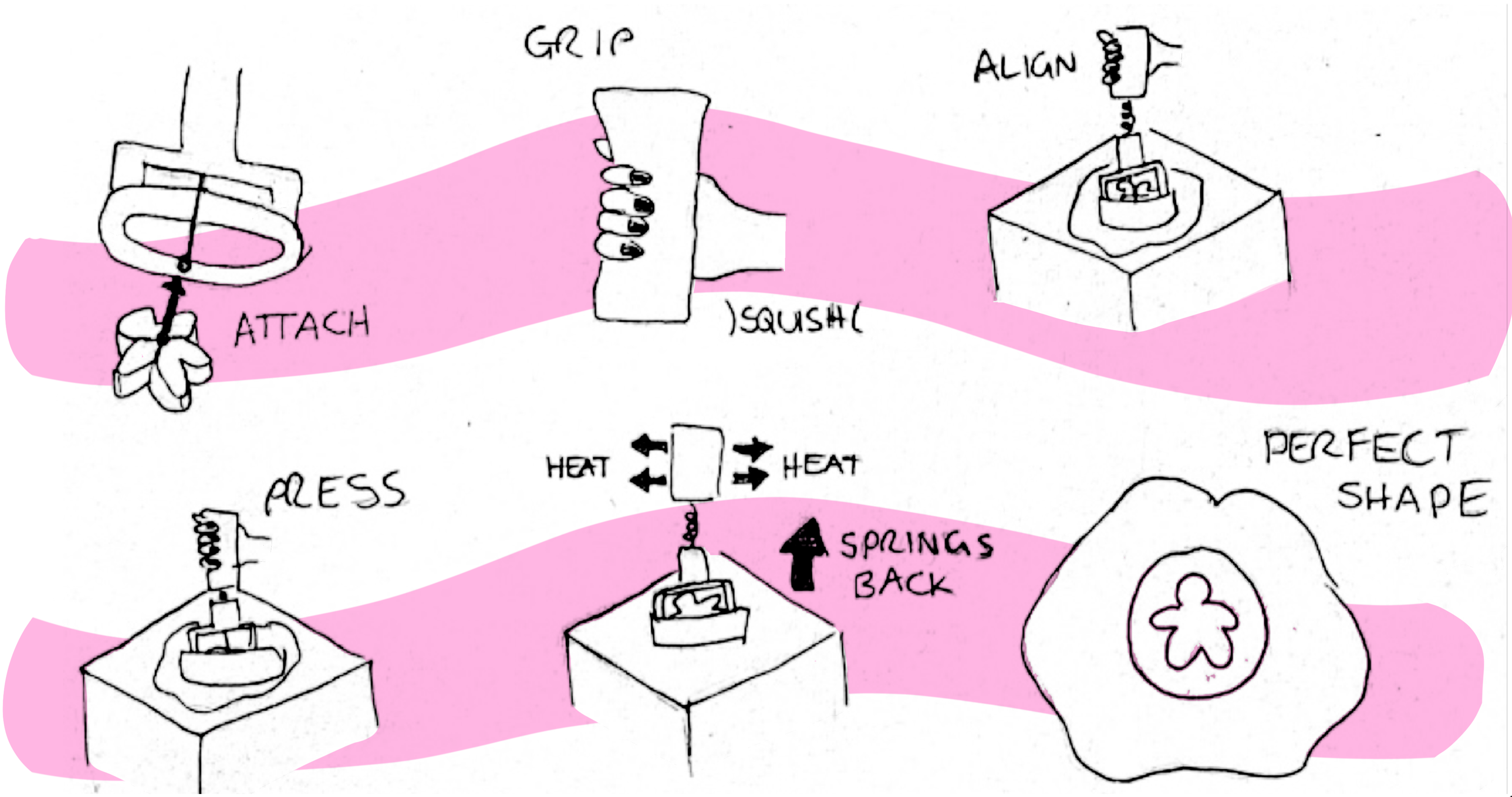
Arthritic hands often have swollen knuckles or joint deformity.





The cookie-cutter shapes can be attached via a magnet connection.

Initial storyboard



There is also a potato masher attachment!

Benchmarking

Reference	Requirement	Accessible Presser	Potato masher for arthritis (A)	Electric knife for arthritis (B)	Crank torch (C)	Analysis
1	Handle is malleable, yet elastic					Though A and B may have comfortable handles, which are designed for users with arthritis, they do not share the malleable quality of my concept
2	Handle must fit comfortably in the user's hand					C has a very small handle.
3	Works with a range of attachments					All other concepts only work with their original set-up.
4	Safety features in place so the user isn't harmed					All concepts have suitable safety measures in place.
5	Rust resistant material					Where metal is present, it is rust resistant.
6	Converts kinetic energy to electrical energy					Neither A nor B repurposes kinetic energy.

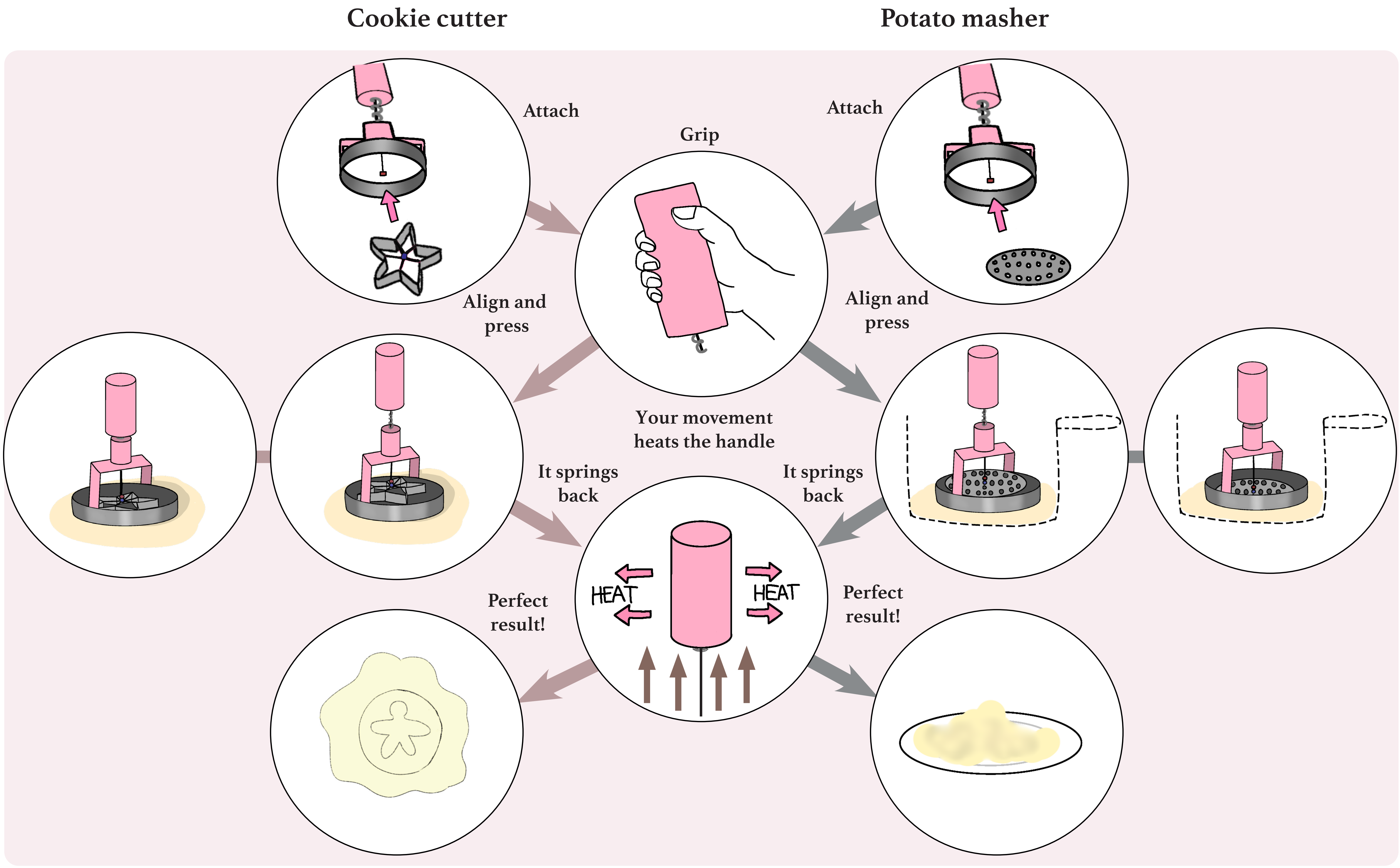
A

B

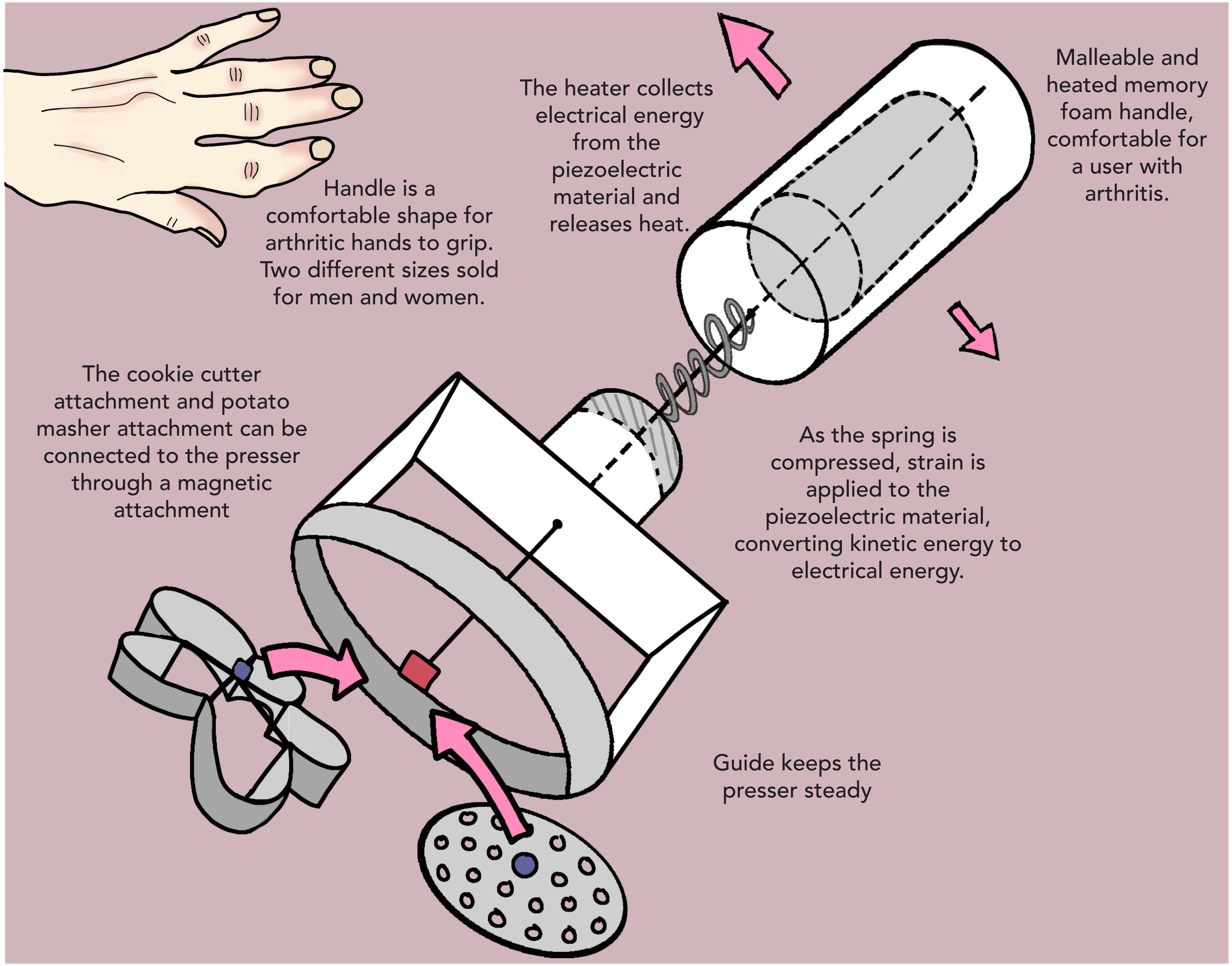
C

Final Storyboard

Choose your attachment



Final design - Accessible Presser



Design Validation and Review

Requirements	Validation
Handle must heat up using re-purposed kinetic energy	Kinetic energy is re-purposed into electrical energy through the piezoelectric material and the pressing down of the handle.
Malleable while elastic handle so it doesn't fracture under repeated reshaping	Handle is made from malleable memory foam.
Handle must correlate to the anthropometric data of palm width.	Two handle sizes produced for both the 95th percentile of men and women.
Pin must attach to a range of cookie-cutter shapes and the potato masher attachment.	Magnetic attachment for easy connection.
Attachments must be safe, metal must not cut flesh	Edges on all metal parts are rounded.
Rust resistant spring	Zinc coating on the spring to prevent rusting.

Third party input

I interviewed two people with arthritis on their opinions on this product:

"Its a nice idea, I do struggle sometimes when I cook because my joints hurt"

- Patricia -

"I like that different bits attach to it. You'll only need the handle then."

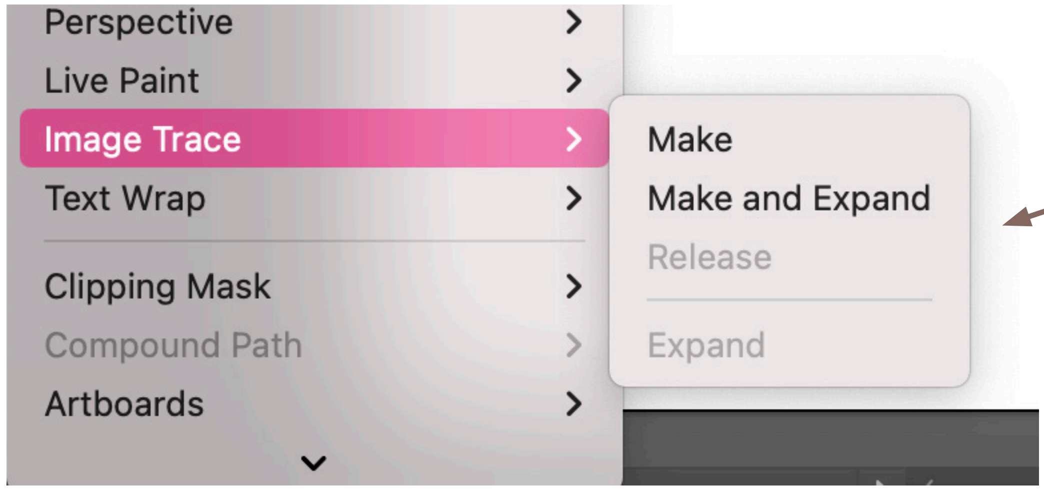
- Julie -

"It recycles energy as well? That's useful."

- Julie -

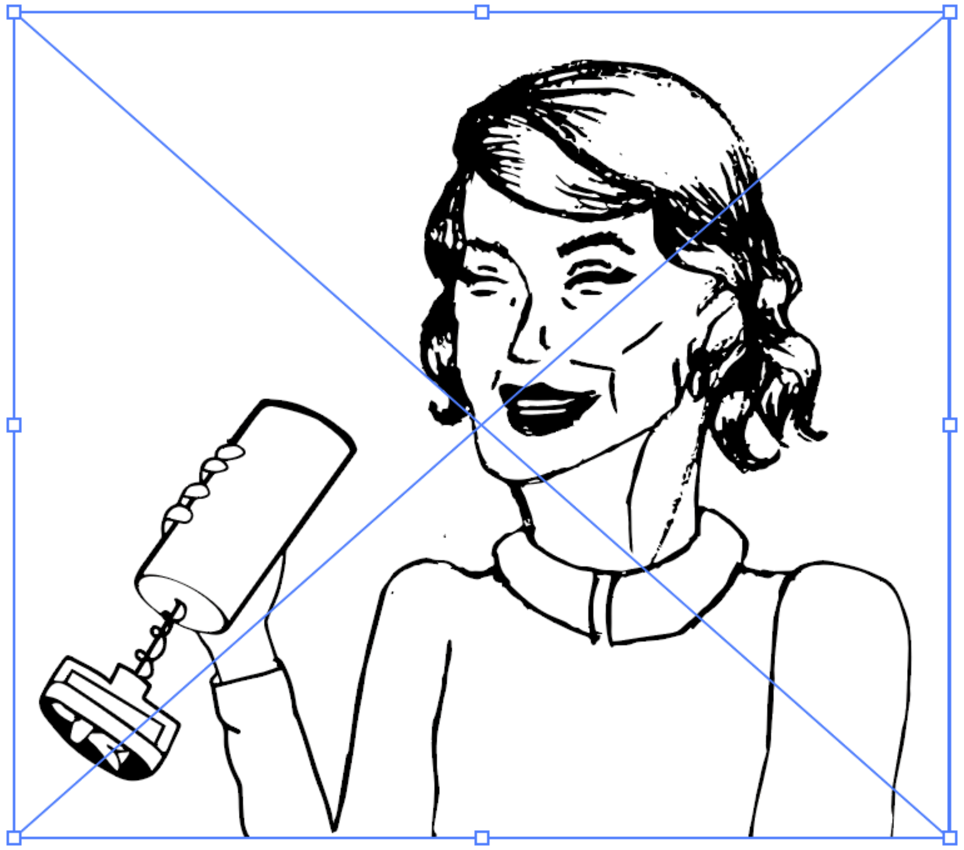
Supplementary Pages

Illustrator exploration



Using image trace in Illustrator

Vector image trace of the sketch



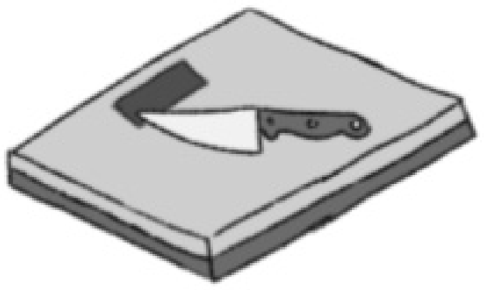
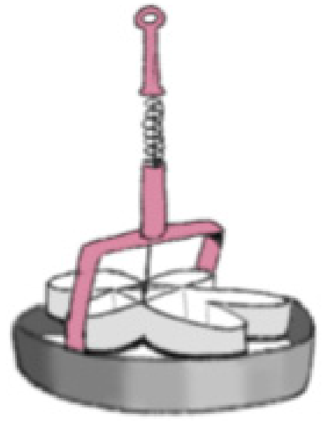
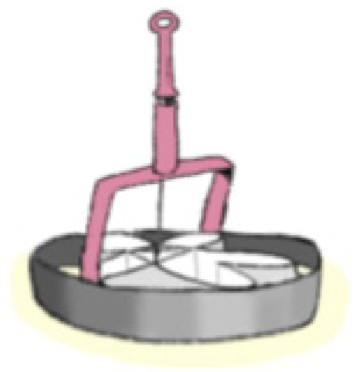
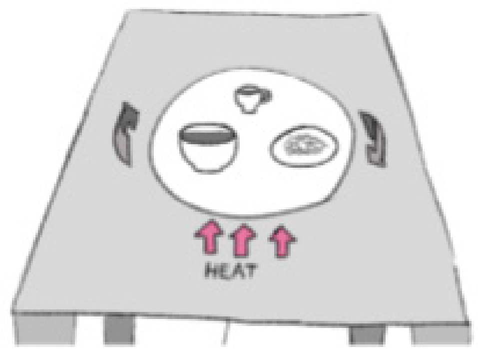
Colouring the image

Tool	Shortcut
Selection	V
Direct Selection	A
Magic Wand	Y
Lasso	Q
Pen	P
Curvature Tool	Shift + ~
Type	T
Touch Type	Shift + T
Line Segment	\
Anchor Point	Shift + C
Add Anchor Point	=
Delete Anchor Point	-
Rectangle	M
Ellipse	L
Paintbrush	B
Blob Brush	Shift + B
Pencil	N
Shaper Tool	Shift + N
Scissors	C
Rotate	R
Reflect	O
Free Transform	E
Perspective Grid	Shift + P
Perspective Selection	Shift + V
Warp	Shift + R
Width	Shift + W
Eraser	Shift + E
Mesh	U
Gradient	G
Eyedropper	I
Blend	W
Scale	S
Column Graph	J
Shape Builder	Shift + M
Live Paint Bucket	K
Live Paint Selection	Shift + L
Artboard	Shift + O
Slice	Shift + K
Hand	H
Zoom	Z
Color	,
Gradient	.
Default	D
None	/
Toggle Fill/Stroke	X
Swap Fill/Stroke	Shift + X
Symbol Sprayer	Shift + S
Toggle Screen Mode	F

Action	Shortcut
Create a document	Command + N
Create a document from a template	Shift + Command + N
Open a document	Ctrl + O
Place a file in the document	Shift + Command + P
Open the Export for screens dialog box	Option + Command + E
Open the Save For Web dialog box	Option+Shift+Command+S
Package the document	Option+Shift+Command+P
Open the File Information dialog box	Option+Shift+Command+I
Print	Command + P
Exit the application	Command + Q
Open the Color Settings dialog box	Shift + Command + K
Open the Preferences dialog box	Command + K
Repeat transforming objects in perspective	Command + D
Move an object	Shift + Command + M
Group the selected artwork	Command + G
Ungroup the selected artwork	Shift + Command + G
Make a clipping mask	Command + 7
Select artwork in active artboard	Command + Option + A
Deselect	Shift + Command + A
Reselect	Command + 6
Select the object above the current selection	Option + Command +]
Select the object below the current selection	Option + Command + [
Make Live Paint (when using the Paint Bucket tool)	Option + Command + X
Zoom in	Command + =
Zoom out	Command + -
View all artboards in window	Command + 0 (zero)
Show/ hide artboard rulers	Command + R
Show/ hide smart guides	Command + U
Show grid	Command + '
Show/ hide Align panel	Shift + F7
Show/ hide Appearance panel	Shift + F6
Show/ hide Color panel	F6
Show/ hide Gradient panel	Command + F9
Show/ hide Graphic Styles panel	Shift + F5
Show/ hide Info panel	Command + F8
Show/ hide Layers panel	F7
Show/ hide Stroke panel	Command + F10
Show/ hide Symbols panel	Shift + Command + F11
Open the Character panel	Command + T
Open the Paragraph panel	Option + Command + T
Show/ hide Transform panel	Shift + F8
Show/ hide Pathfinder panel	Shift + Command + F9
Add new fill	Command + /
Add new stroke	Command + Option
Add a layer	Command + L
Add a layer while opening the New Layer dialog box	Option + Command + L

Exploring Illustrator short-cuts

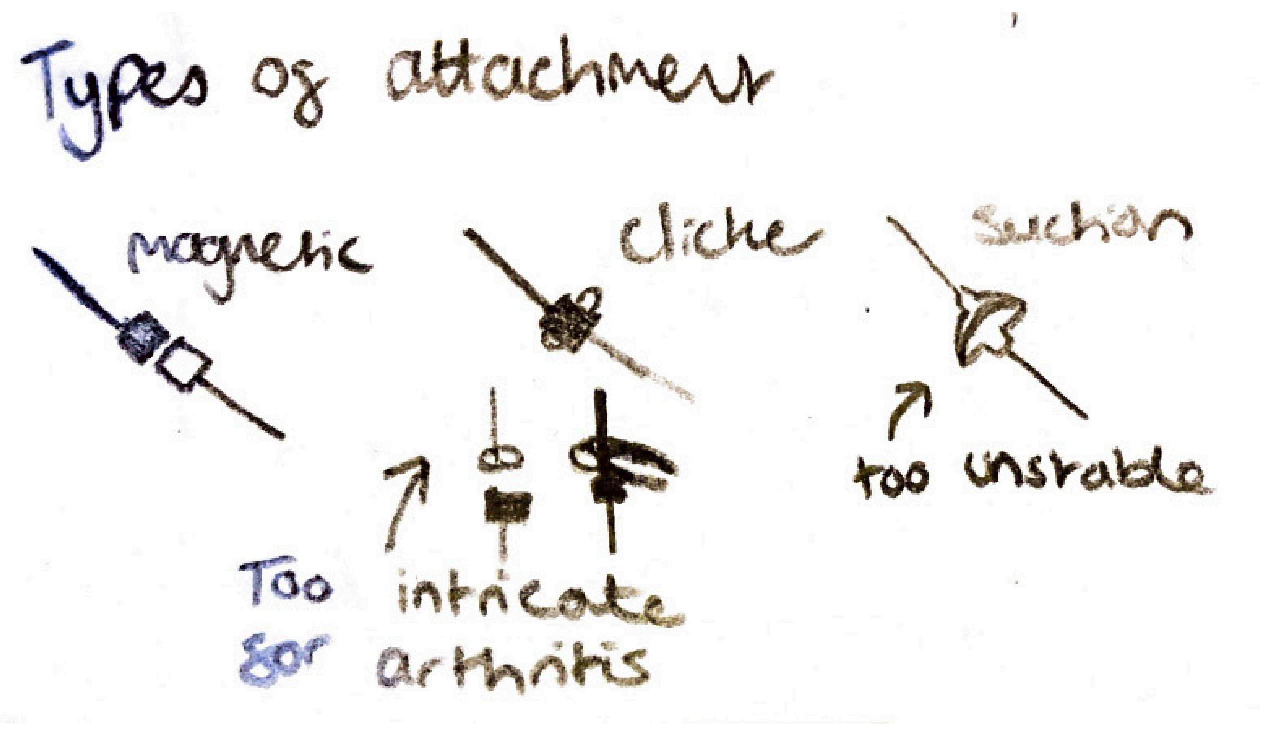
Group requirements evaluation of Part A ideas

	A	★ B*	C 😀*
Title	Chopping board weighing scale	Spring-loaded cookie cutter	Heated Lazy Suzan
HMW	Repurpose KE to improve user experience	<-	<-
Requirement 1	Fit comfortable on a countertop Easy to clean	Malleable while elastic handle so it doesn't fracture under repeated reshaping Rust resistant parts (especially spring as steels are commonly used to make them)	Waterproof in case of spills Could have a boarder around it so that it is not hot to touch whilst rotating, for this consider width of lazy susan and length of boarder to match the typical griping methods of users
Requirement 2	Fit a large knife Hard coating (scratch resistant)	Spring is stiff enough to overcome stickiness of dough but not too stiff to freak out the user when it returns to original position Quantify this? Maybe Youngs modulus of material Handle much corelate to the anthropometric data of average palm width (consider 95th percentile)	Knows when to stop heating so that the surface only gets warm but not hot Temperature should be between (x) degrees and (y) degrees Diameter of lazy susan must be related to the diameter of the table, typically the lazy susans exist on circular tables, could the disk shape be altered so that It fits other table shapes
Requirement 3	Be able to display reading for a certain period Relatively lightweight as it seems like this product would be one you would bring out from storage to use rather than have present always due to it being quite large. Alternatively you could consider methods of incorporating it into the counter-top	Suitable safety mechanisms in place? Cutter must not be too sharp that it can cut flesh Could add extra parts to be sold sepeartely to create different shaped cookies	Mechanism to stop the outside of the Suzan that you spin from getting too hot Method of allowing the user to grip the lazy susan, maybe some indents or handles at regular points around
	Durable protective layer (e.g. tempered glass/ceramic coating/sapphire glass) for display to avoid damage from knife		
	Surface: weight-sensitive while being sturdy (why are scales usually made of glass?)		
	Knife must not be able to cut through board and damage circuit		

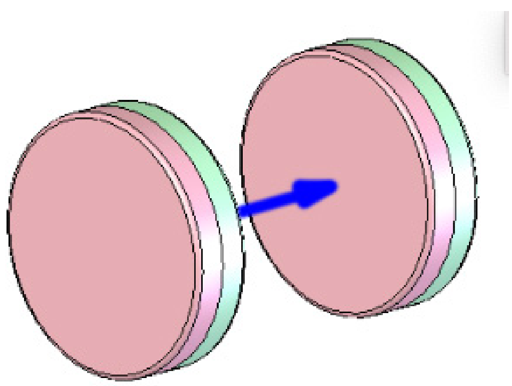
Exploring attachment styles

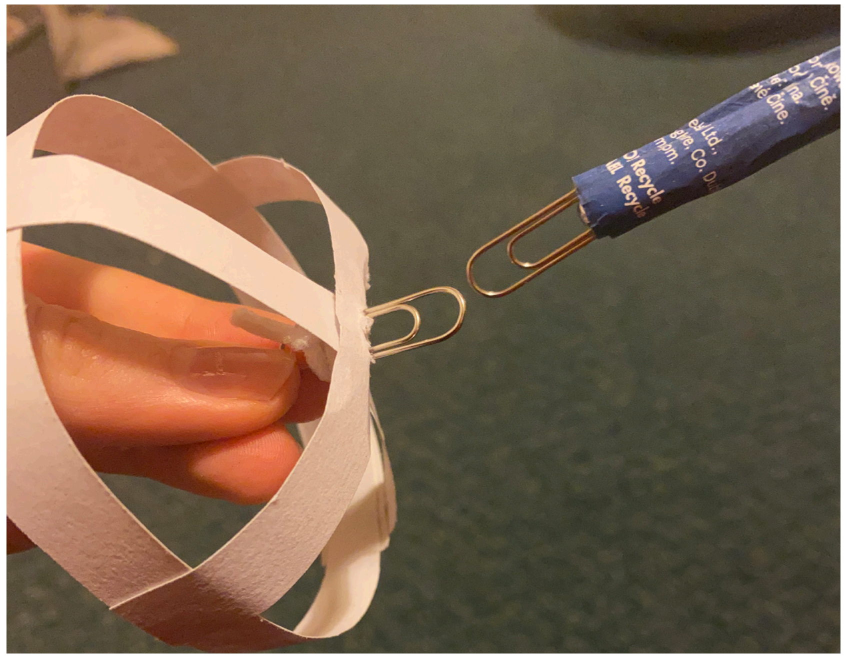
Thinking through sketching
Lofi prototyping

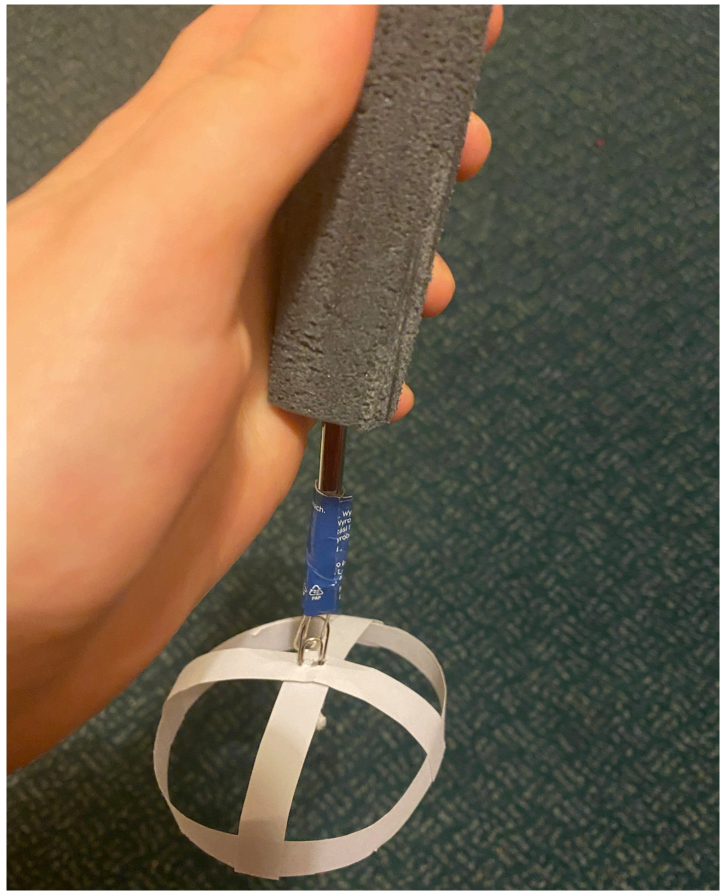


Quick lofi prototyping to determine if the clicker attachment is too intricate for someone with arthritis.

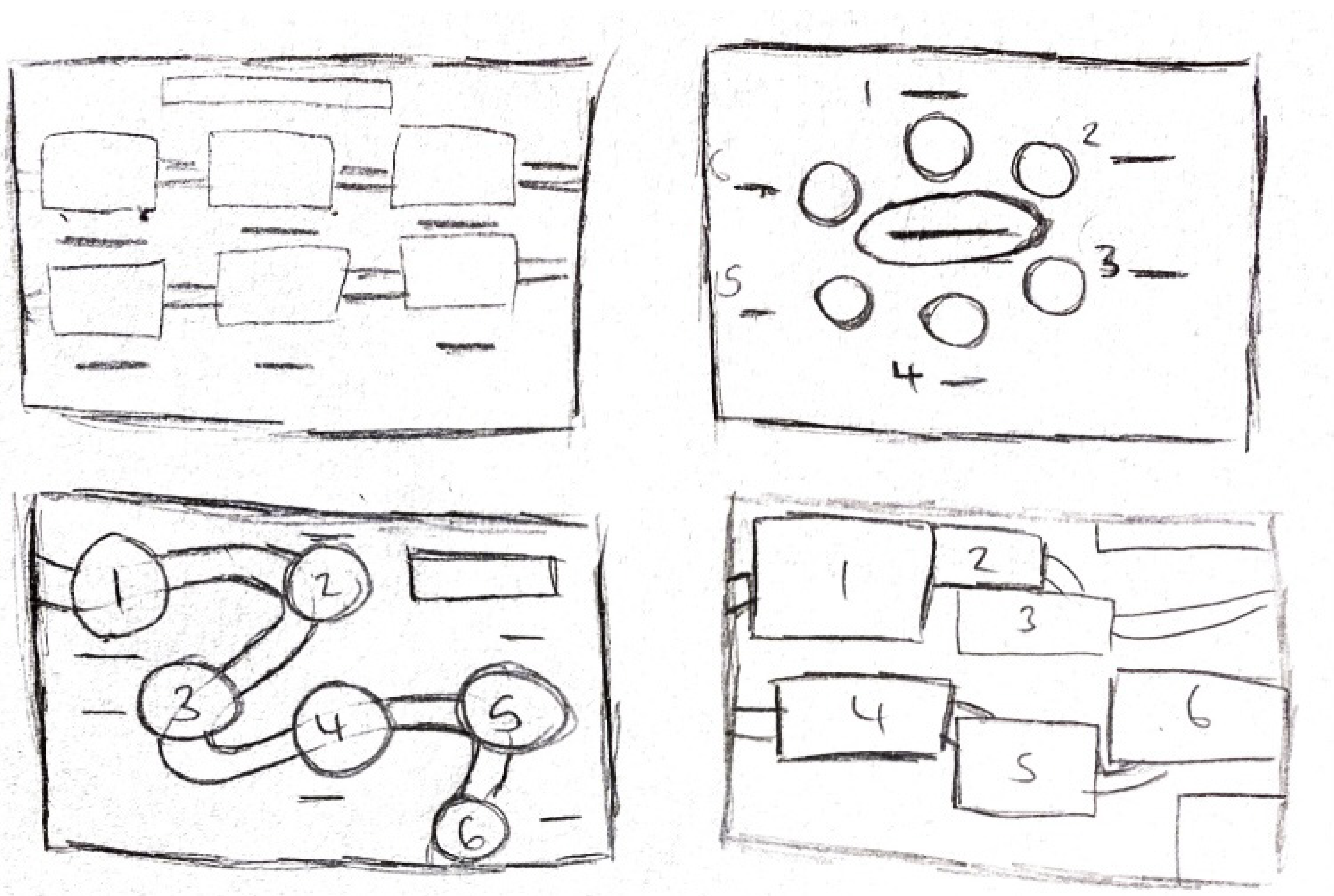
The attachment was modelled using two paper clips. This simulates how the clicker attachment slides and sticks together. The paper clip attachment felt too intricate as a non-arthritic person which means the clicker attachment will definitely be unsuitable.



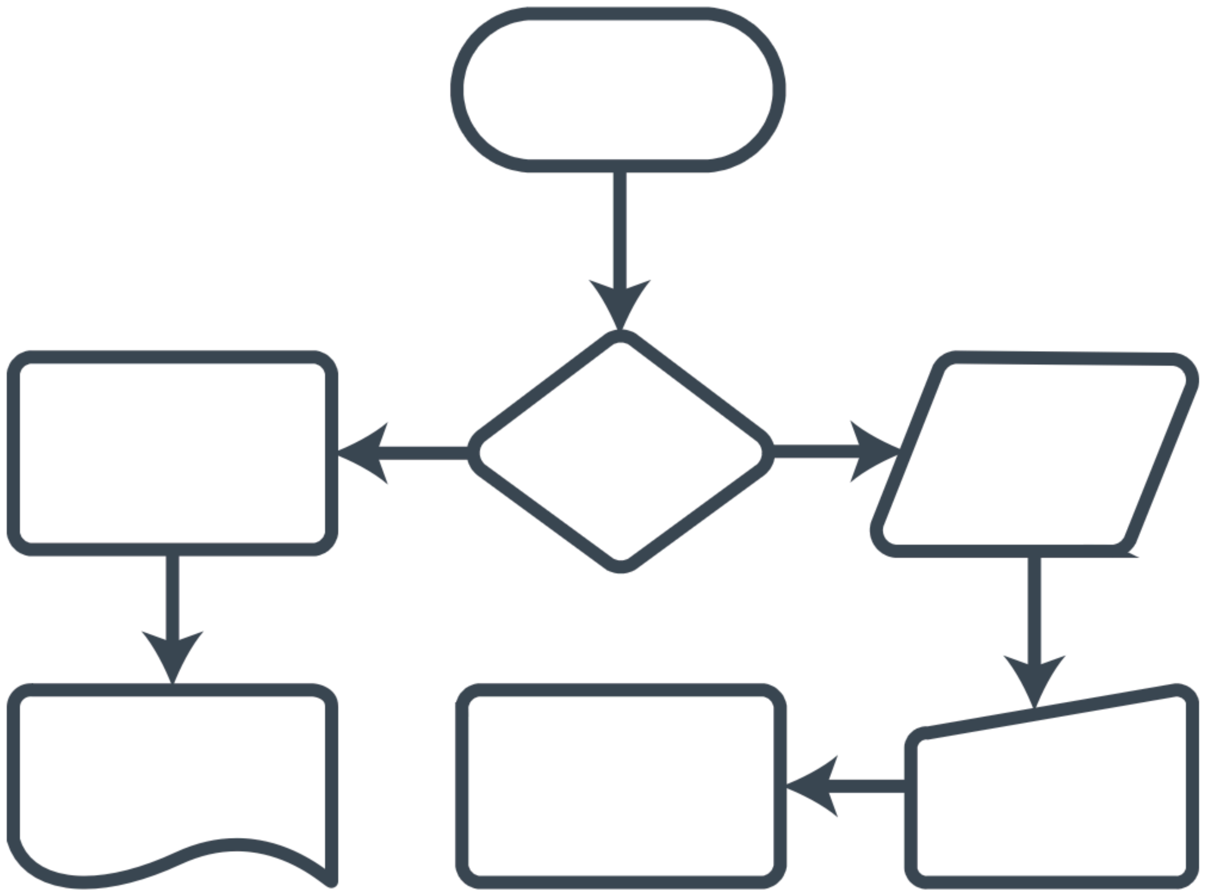




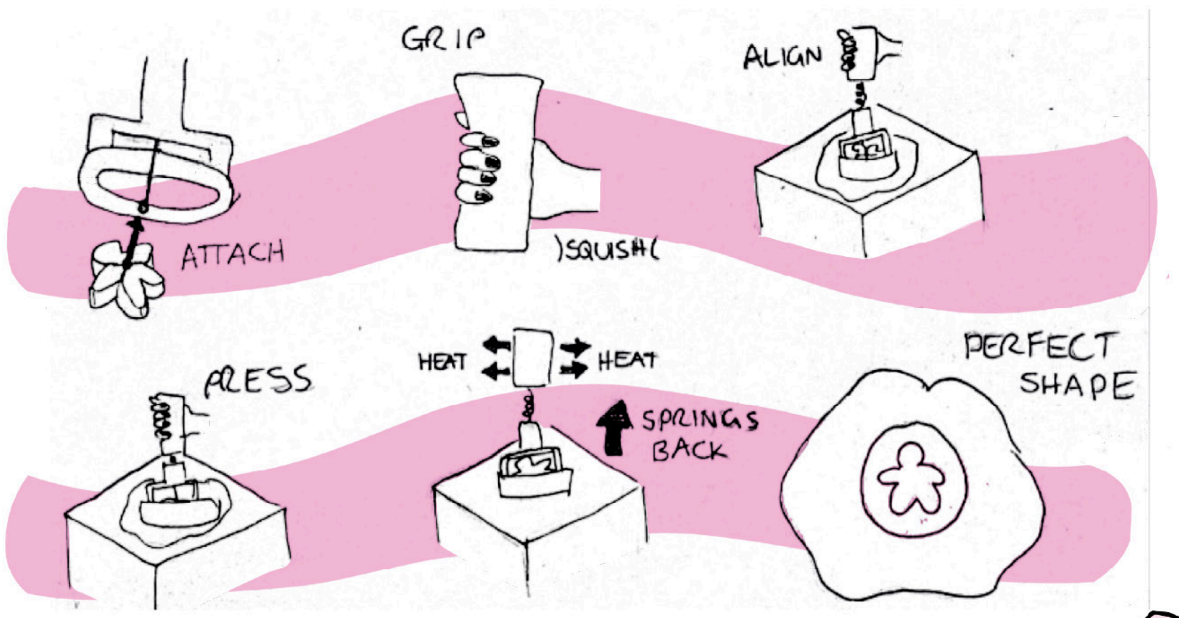
Storyboard layout quick brainstorming

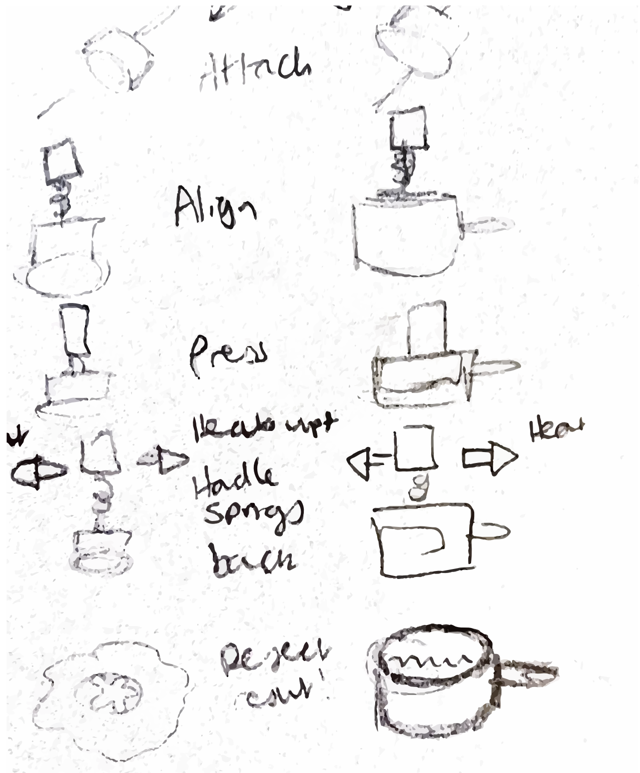


Final storyboard layout brainstorming

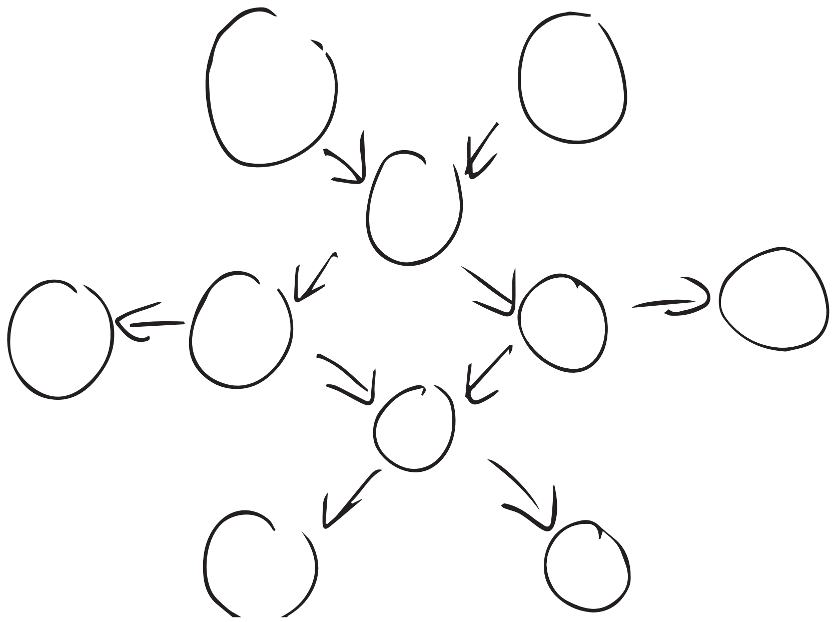


Initial storyboard

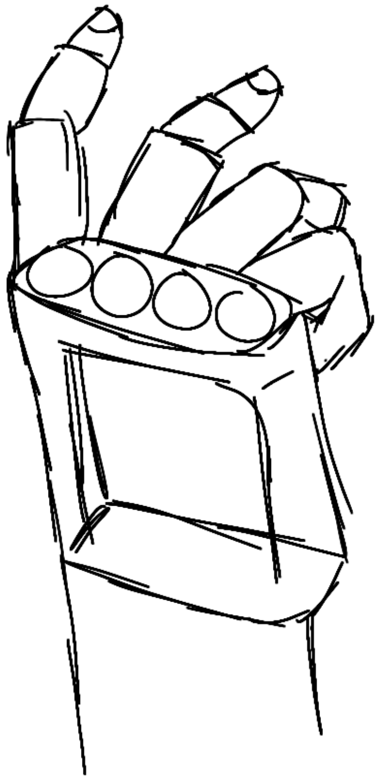


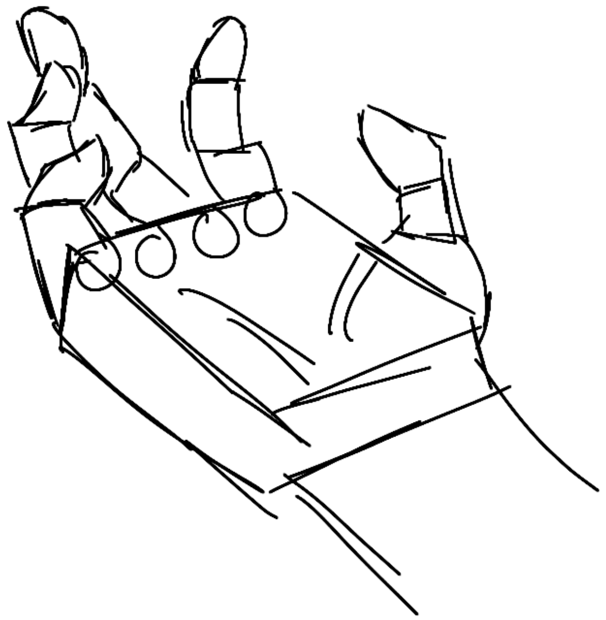


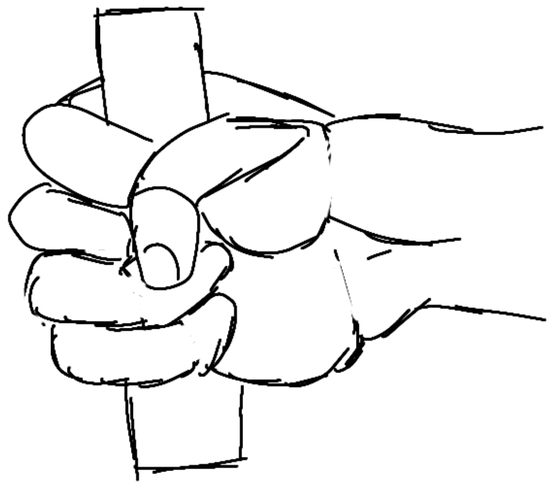
Combine a flowchart with the initial storyboard to showcase both of the different attachments

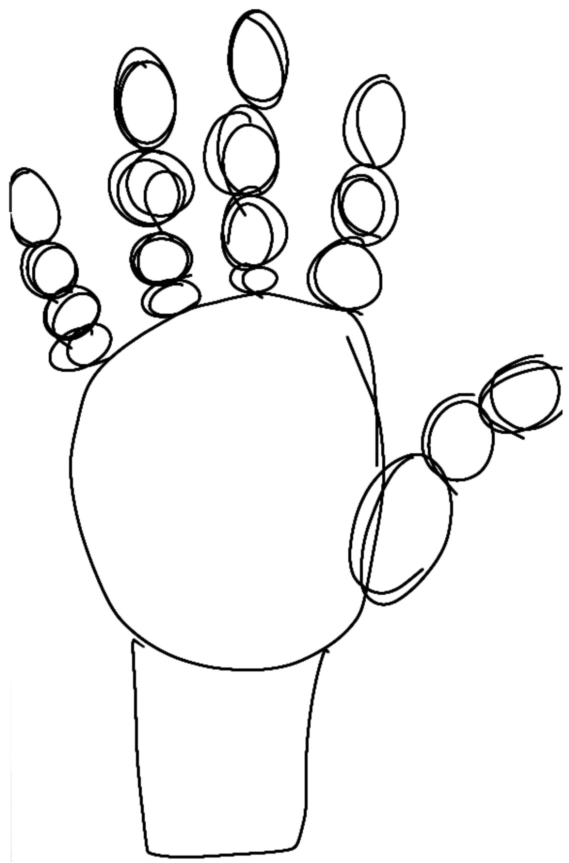


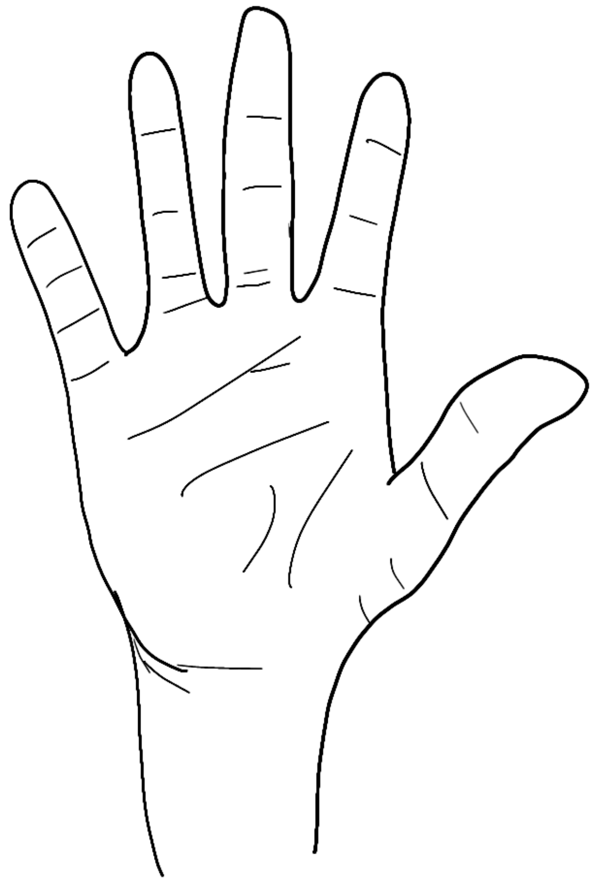
Quick hand sketching



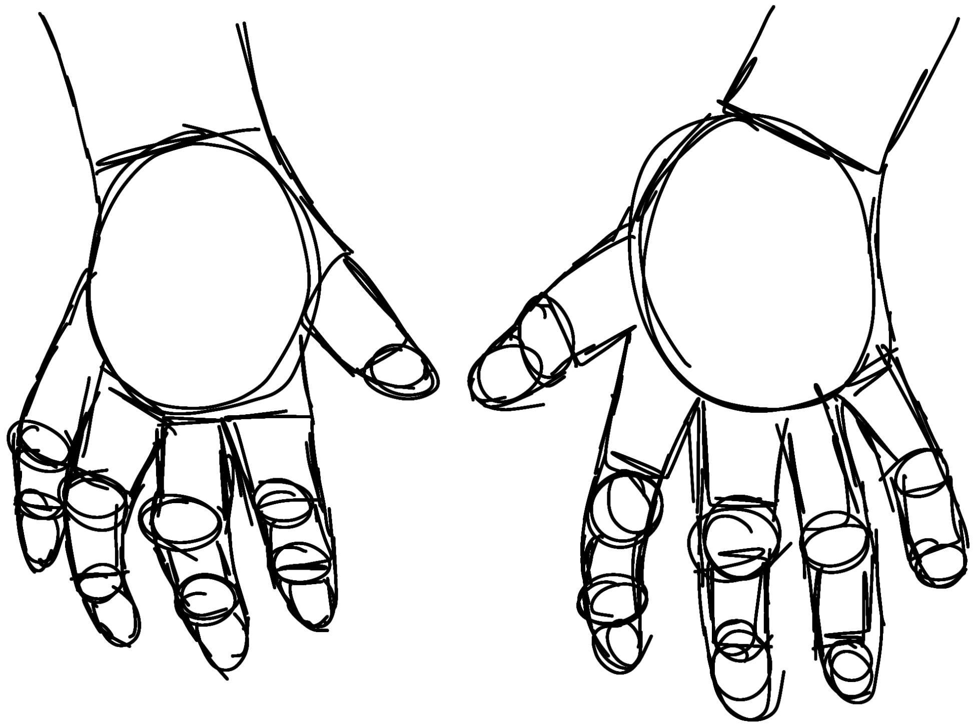


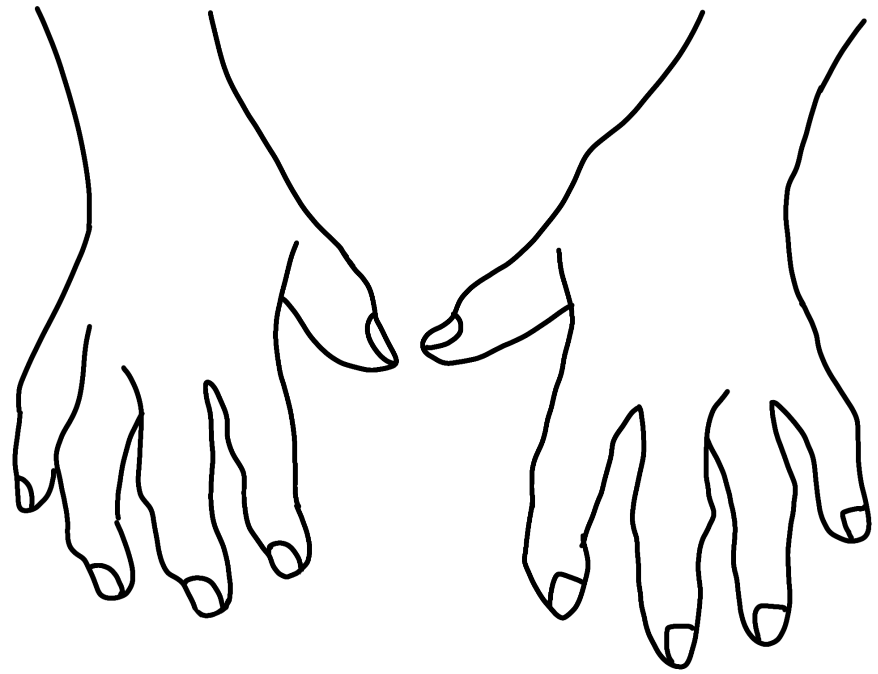


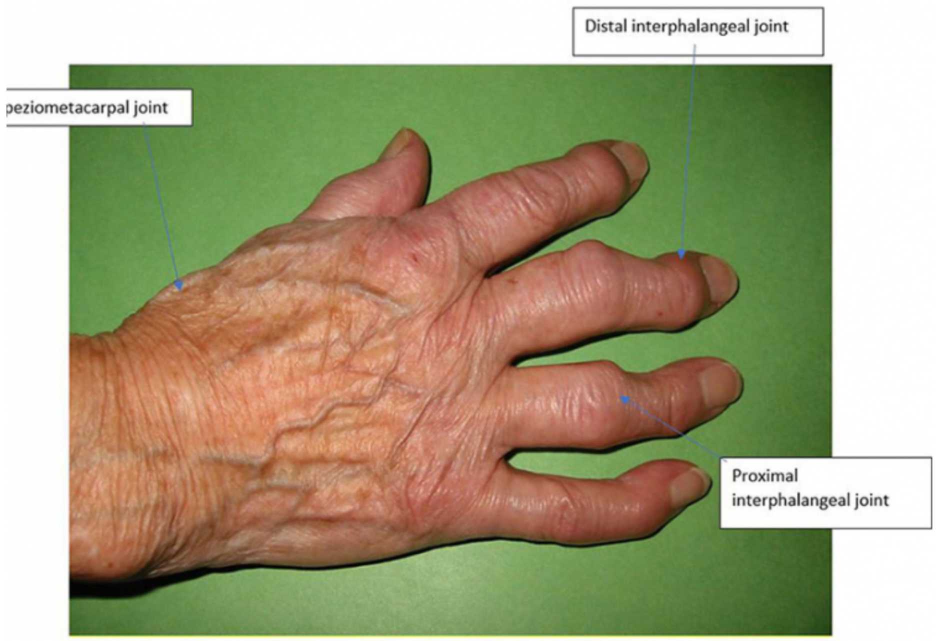


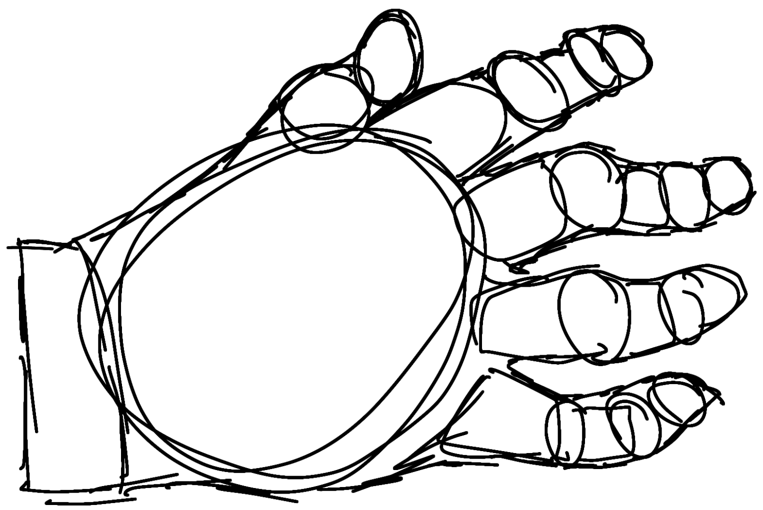


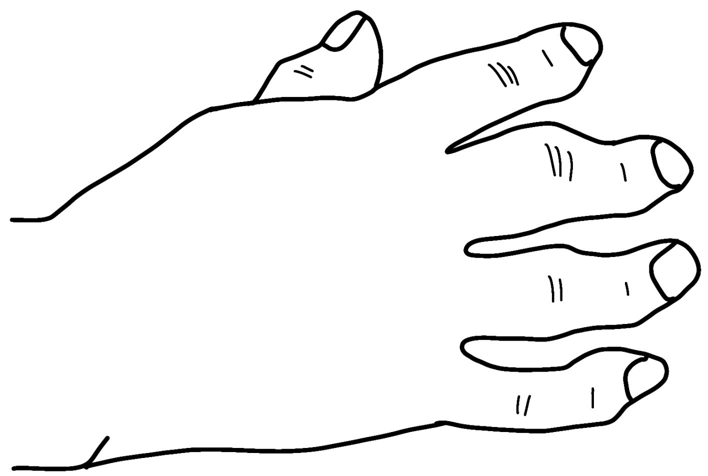
Arthritic hand sketching











Fin.